

Electromagnetism 2

(spring semester 2026)

Lecture 6

Displacement current; Maxwell's equations

- ❖ The continuity equation
- ❖ Generalization of Ampere's law for non-steady currents
- ❖ The displacement current; examples
- ❖ Maxwell's equations of in free space

Previous lectures

- ❖ Our current (still incomplete) understanding of the equations of electromagnetism *in free space*:

Gauss law (*universally valid*)

$$\nabla \vec{E} = \rho / \epsilon_0$$

Faraday's law (*universally valid*)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Absence of magnetic poles
(*universally valid*)

$$\nabla \vec{B} = 0$$

Ampere's law (*static field only*)

$$\nabla \times \vec{B} = \mu_0 \vec{j}$$

- ❖ Variation of magnetic field gives rise to electric field.
- ❖ Does variation of electric field give rise to magnetic field?

The continuity equation

Conservation of electric charge:

any variation of charge within a fixed volume is due to charges flowing across the surface

$$\int_S \vec{j} d\vec{S} = -\frac{d}{dt} \int_V \rho dV$$

Unit on both sides: [C/s=A]

Divergence theorem: $\int_S \vec{j} d\vec{S} = \int_V \nabla \cdot \vec{j} dV$

Conservation of charge in differential form:
(the *continuity equation*)

$$\nabla \cdot \vec{j} = -\frac{\partial \rho}{\partial t}$$

Let's take the divergence of both sides of Ampere's law:

$$\underbrace{0 = \nabla(\nabla \times B)}_{\text{T1 from lecture 2}} = \underbrace{\mu_0 \nabla \cdot \vec{j}}_{\text{Continuity equation}} = -\mu_0 \frac{\partial \rho}{\partial t}$$

Therefore, Ampere's law applies to **steady currents only**
($\partial \rho / \partial t = 0$), and **requires generalisation**

Generalised Ampere's law

To generalise Ampere's law for non-steady currents, let's modify its right-hand side to make its divergence zero.

Gauss law: $\epsilon_0 \nabla \cdot \vec{E} - \rho = 0$. Differentiation over t :

$$0 = \epsilon_0 \nabla \cdot \left(\frac{\partial \vec{E}}{\partial t} \right) - \frac{\partial \rho}{\partial t} = \epsilon_0 \nabla \cdot \left(\frac{\partial \vec{E}}{\partial t} \right) + \nabla \cdot \vec{j} = \nabla \cdot \left(\underbrace{\epsilon_0 \frac{\partial \vec{E}}{\partial t} + \vec{j}}_{\text{Displacement current density}} \right)$$

We have found a quantity $\left(\epsilon_0 \frac{\partial \vec{E}}{\partial t} + \vec{j} \right)$ which always has zero divergence and is therefore suitable for a modified Ampere's law.

The *Ampere-Maxwell law*: $\nabla \times \vec{B} = \mu_0 \vec{j} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} \quad (*)$

Not a proof: there are many ways of removing the contradiction. Any solenoidal field (i.e. such that $\nabla \cdot \vec{u} = 0$) can be added to the right-hand side of (*), still satisfying charge conservation.

The displacement current

$$\nabla \times \vec{B} = \mu_0(\vec{j}_C + \vec{j}_D) = \mu_0\vec{j}_C + \epsilon_0\mu_0 \frac{\partial \vec{E}}{\partial t}$$

$\vec{j}_C$: conduction current density $\vec{j}_D$: displacement current density

A changing **E**-field produces **B**-field:
Maxwell's decisive step towards electrodynamics (**1865**).

It took **30** years after Faraday's discovery of EM induction to postulate the displacement current as a source of magnetic field.

Alternating current in a copper wire [conductivity $\sigma = 6 \times 10^7 (\Omega \cdot m)^{-1}$]

Electric field:

$$E = E_0 \sin(\omega t)$$

Conduction current density:

$$j_C = \sigma E = \sigma E_0 \sin(\omega t)$$

Displacement current density:

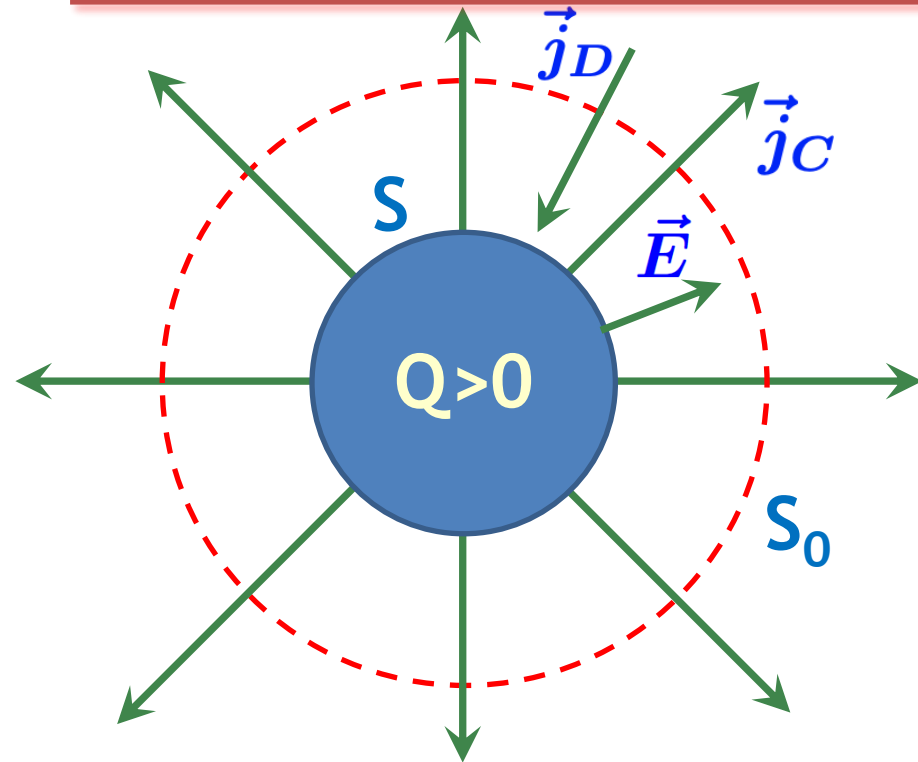
$$j_D = \epsilon_0 \partial E / \partial t = \epsilon_0 \omega E_0 \cos(\omega t)$$

The ratio of maximum max conduction to max displacement current:

$$j_C^{\max} / j_D^{\max} = \sigma E_0 / \epsilon_0 \omega E_0 \sim 10^{19} s^{-1} \omega^{-1} \gg 1,$$

i.e. j_D is negligible for all frequencies used in practice.

Example 1



A charged sphere discharging into an external conductive medium.

By symmetry, $\vec{j}_C$ is directed radially. What about the $\vec{B}$ field?

By symmetry, tangential component $B_t = 0$, and radial component B_R is the same in each point of sphere S .

Assuming $B_R \neq 0$ leads to $\int_S \vec{B} d\vec{S} \neq 0$

We conclude that $\vec{B} = 0$.

Ampere's law leads to $\vec{j}_C = 0$.

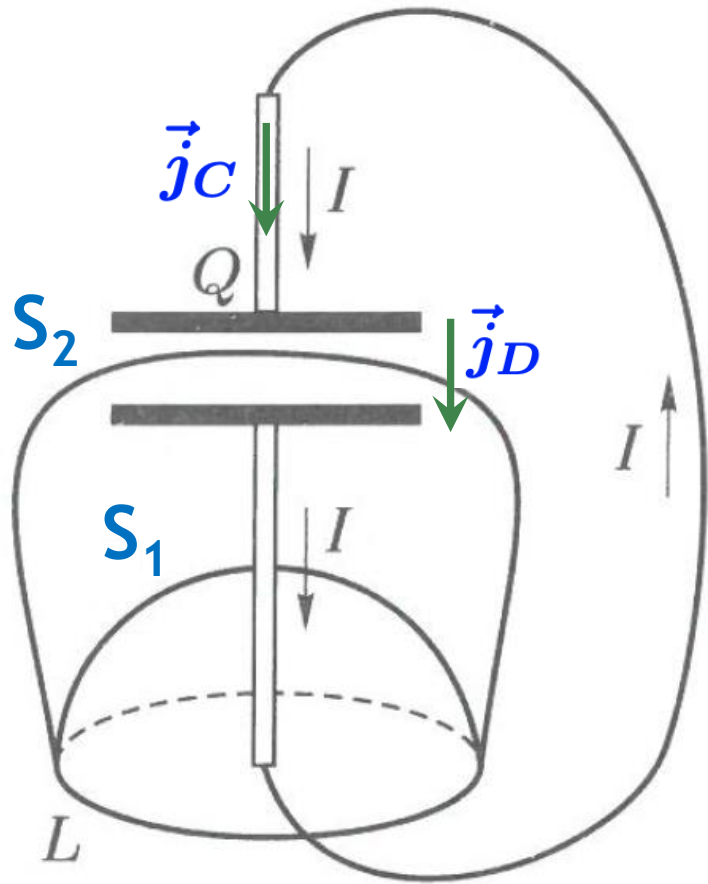
$\vec{E}$ -field at the surface of the sphere (*lecture 1*): $E = Q / (4\pi\epsilon_0 R^2)$

Conduction current at surface S_0 is balanced by displacement current:

$$I_D = I_C = \frac{dQ}{dt} = 4\pi\epsilon_0 R^2 \frac{\partial E}{\partial t}$$

$$J_D = I_D / (4\pi R^2) = \epsilon_0 \frac{\partial E}{\partial t}$$

Example 2



Discharge of a parallel-plate capacitor:

$$\oint_L \vec{B} d\vec{l} = \mu_0 \int_S (\vec{j}_C + \vec{j}_D) d\vec{S}$$

- ❖ For the surface S_1 , only the conduction current I_C contributes.
- ❖ For the surface S_2 , only the displacement current I_D contributes.

Let's check explicitly the equality $I_D = I_C$:

$$I_C = \frac{\partial Q}{\partial t} = A \frac{\partial \sigma}{\partial t} = A \epsilon_0 \frac{\partial E}{\partial t}$$

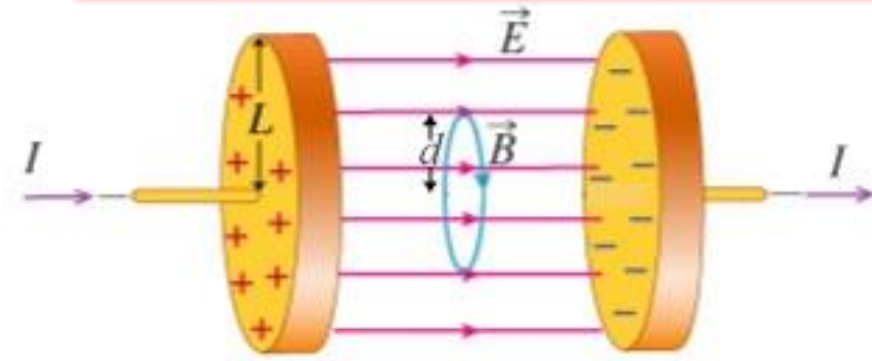
By definition, $I_D = A \epsilon_0 \frac{\partial E}{\partial t} = I_C$

Lecture 1: $E = \sigma / \epsilon_0$

[A: area of the plates]

Example 3

[not discussed in the lecture]



A thin parallel plate capacitor with circular plates of radius R is being charged. Find the magnetic field $\mathbf{B}(\mathbf{r})$.

Using axial symmetry, in the absence of conduction current ($\mathbf{j}_C = \mathbf{0}$),

$$\oint_L \vec{B} d\vec{l} = \mu_0 \int_S (\vec{j}_C + \vec{j}_D) d\vec{S} = \epsilon_0 \mu_0 \int_S \frac{d\vec{E}}{dt}$$

Inside the capacitor ($r < R$),

$$2\pi r B = \epsilon_0 \mu_0 \cdot \pi r^2 \frac{dE}{dt}$$

$$B(r) = \frac{1}{2} \epsilon_0 \mu_0 r \frac{dE}{dt}$$

Outside the capacitor ($r \geq R$),

$$2\pi r B = \epsilon_0 \mu_0 \cdot \pi R^2 \frac{dE}{dt}$$

$$B(r) = \frac{1}{2} \epsilon_0 \mu_0 \frac{R^2}{r} \frac{dE}{dt}$$

Maxwell's equations in free space

Laws of electrodynamics in free space:

(remember them)

$$(M1) \quad \nabla \vec{E} = \rho / \epsilon_0$$

$$(M2) \quad \nabla \vec{B} = 0$$

$$(M3) \quad \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$(M4) \quad \nabla \times \vec{B} = \mu_0 \vec{j} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

Displacement current density: $\vec{j}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

The continuity equation, $\nabla \cdot \vec{j} = - \frac{\partial \rho}{\partial t}$, follows from (M1) and (M4).

Discussion and summary

- ❖ Maxwell's equations in free space (**2** scalar + **2** vector) are equivalent to **8** scalar equations with **10** variables ($E_x, E_y, E_z, B_x, B_y, B_z, j_x, j_y, j_z, \rho$).
- ❖ Maxwell's equations must be complemented by the equations characterising the media. These will be discussed in lectures 7–12.
- ❖ The equations are not symmetric wrt electric and magnetic fields:
 - ✓ no magnetic poles as sources of **B** field (**M2**);
 - ✓ no magnetic currents as sources of **E** field (**M3**).
- ❖ For constant fields ($\partial E/\partial t = \partial B/\partial t = 0$), two independent groups,
 - ✓ electrostatics: $\text{div } E = \rho/\epsilon_0$; $\text{curl } E = 0$;
 - ✓ magnetostatics: $\text{div } B = 0$; $\text{curl } B = \mu_0 j$.